

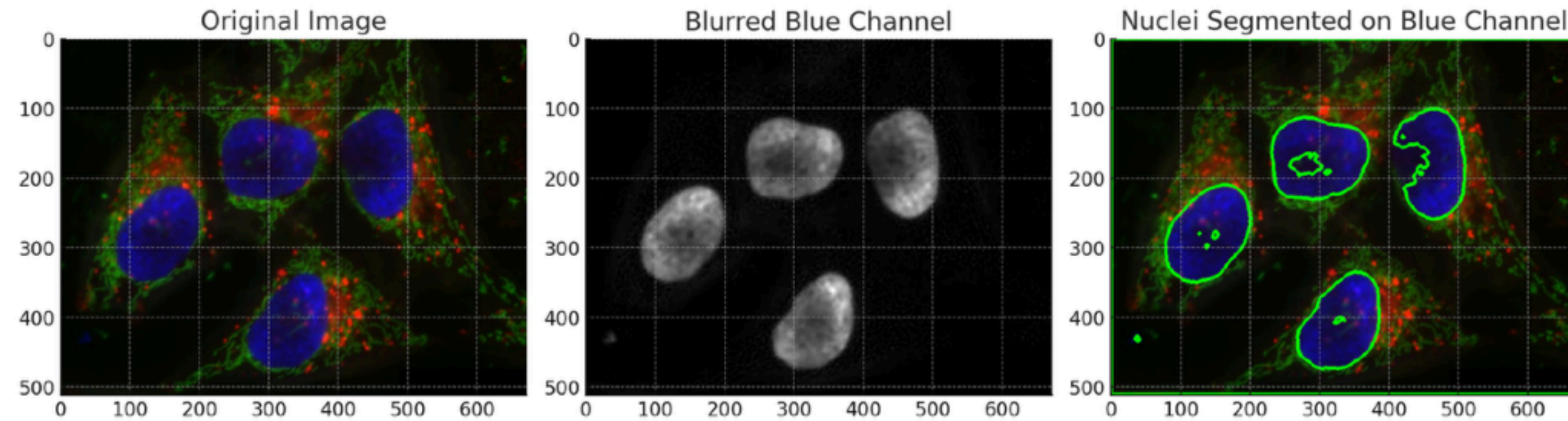
Using ChatGPT-4 for Image Analysis

```
python Copy code  
  
import cv2  
import numpy as np  
from matplotlib import pyplot as plt  
  
# Load the image  
image_path = '/mnt/data/hela-cells.jpg'  
image = cv2.imread(image_path)  
  
# Convert the image to gray scale  
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)  
  
# Apply GaussianBlur, which is good for edge detection  
blurred = cv2.GaussianBlur(gray, (5, 5), 0)  
  
# Perform Otsu's thresholding  
_, thresh = cv2.threshold(blurred, 0, 255, cv2.THRESH_BINARY_INV + cv2.  
  
# Find contours which will give us the nuclei boundaries  
contours, _ = cv2.findContours(thresh, cv2.RETR_TREE, cv2.CHAIN_APPROX_  
  
# Draw the contours on the original image  
contoured_image = image.copy()  
cv2.drawContours(contoured_image, contours, -1, (0, 255, 0), 3)  
  
# Save the output image with segmented nuclei  
output_path = '/mnt/data/hela-cells-segmented.jpg'  
cv2.imwrite(output_path, contoured_image)
```

Using ChatGPT-4 for Image Analysis

You
please use the blue channel for segmenting the nuclei

ChatGPT



The nuclei segmentation has been performed using the blue channel of the image with the following steps:

- The blue channel was extracted from the original image since the nuclei appear to be stained blue.
- Gaussian Blur was applied to the blue channel to smooth the image, which helps in reducing noise for thresholding.
- Otsu's thresholding was then used on the blurred blue channel to create a binary image where the nuclei are distinguished from the background.